Sea-land Segmentation of Infrared Remote Sensing Image Based on Complex Background

Baorong Xie*
Shanghai Aerospace Electronic Technology Institute, Shanghai, China
021-24181944,201109
littlecrab1024@126.com

Ye Heng
Shanghai Aerospace Electronic Technology Institute, Shanghai, China
021-24182089,201109
Yeheng09@163.com

Shuyi Feng
Shanghai Aerospace Electronic Technology Institute, Shanghai, China
021-24181944,201109
feng_shu_yi@aiyun.com

Yang Wang
Avic Atm System& Equipment Co. Ltd, Shanghai, China
021-24253132,200241
wang_yang@careri.com

Xinzhong Zhu
Shanghai Aerospace Electronic Technology Institute, Shanghai, China
021-24181944,201109
zhuxinzhong@yeah.net

***Abstract*-With the development of infrared payload technology, the resolution is gradually improved, which makes it possible for infrared payload to detect marine ship targets all-weather. However, due to the influence of atmosphere, ocean thermal radiation, detector noise and other factors, ship targets appear as small targets with low SNR in infrared image. How to recognize the infrared ship in the complex background is a difficult problem. For the infrared ship target, the land and the sea need to be segmented first. For the situation in which the sea-land boundary is not obvious, for example, when the coastal area is covered by clouds or the water depth is shallow, it is difficult to use the traditional methods such as region growing method to realize the sea-land segmentation. For the sea-land segmentation of complex background mentioned above, this paper proposes a real-time positioning and matching method based on blocks combined with power spectrum statistics, regional growth and GIS database template, which realizes the sea-land segmentation of satellite infrared image with low SNR and low contrast.**

Keywords: sea-land segmentation, complex background, GIS template, low SNR, infrared ship detection

I. INTRODUCTION

Due to the characteristics of all-weather, imperceptive and ability of strong cloud penetration, infrared payload has unique performance in the field of remote sensing application, especially in the civil fields or military applications such as target detection, precise guidance, night navigation [1]. At present, the technology of ship identification based on SAR and visible light payload is relatively mature, and the development of ship identification based on infrared payload relatively lags behind [2]. With the development of infrared payload technology, it is possible to detect all-weather marine ship targets with gradually improved resolution. Now the research on ship target recognition is more and more important, which becomes a hot topic in the field of remote sensing image processing and pattern recognition [3]. As the infrared sensor is easily affected by atmosphere, ocean thermal radiation, detector noise and other factors, which makes the ship targets appear as small targets with low signal-to-noise ratio in the infrared image. So it requires that the infrared ship identification technology can be used in a variety of complex background, which can overcome the influence of various factors and has better real-time performance and higher detection rate. [4] It is an urgent problem to be solved in the infrared ship target recognition technology, while sea-land segmentation is the basis of ocean ship target extraction. For infrared image, the contrast of ship target to background is low, so an effective segmentation method needs to be designed. Due to the low resolution of the onboard infrared sensor, the gray value and texture features of the area near the coast are not obvious, especially in the water area surrounded by land. In addition, the epicontinental sea area is vulnerable to the interference of shadows and clouds, so it is difficult to extract the complete coastline contour. The traditional sea-land segmentation method is not effective for complex topography or complex sea surface conditions. Therefore, this paper proposes a sea-land segmentation method based on complex background. On the one hand, the gray-scale characteristics of the image texture combined with the method of power spectrum statistics based on blocks and region growth is used for sea-land segmentation; on the other hand, the method of on orbit positioning solution combined with GIS template database matching is further used to eliminate the land interference area.

978-1-7281-9481-3/20/$31.00 ©2020 IEEE
DOI 10.1109/CVIDL51233.2020.00039

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:39:49 UTC from IEEE Xplore. Restrictions apply.

II. SEA-LAND SEGMENTATION PROCESS

As near the coastline there is no obvious difference in the gray level and texture features and there is interference of ship shadow and occlusion for the infrared remote sensing images, it is difficult to extract the complete coastline contour by using the traditional threshold segmentation methods such as region growing. In this paper, we employ the way of statistics of image power spectrum in the block area. First the conditions of the images are divided into three cases: smooth, complex and mixed. For the smooth case, the block area is considered as sea. When the power spectrum value of the image is relatively high, the block area is determined as land. In the mixed case, the seed points are calculated for the region growing for the initial sea-land separation. However, when the coastal boundary is sheltered by clouds or intertidal zone appearing near the sea, the difference between the sea and land is small, which leads to the results of inaccuracy for the preliminary separation of the sea and land. So the GIS template database is employed further to match the sea land blocks, so as to improve the accuracy of the separation. The specific method is shown in the figure below.

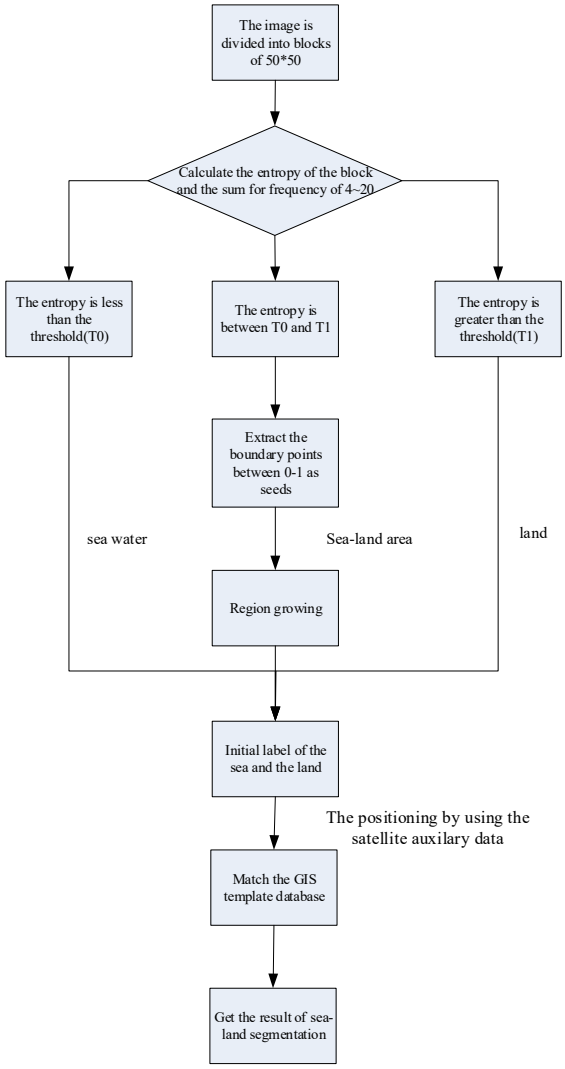


Figure 1. Sea-land segmentation method

III. BLOCK POWER SPECTRUM STATISTICS

Firstly, the statistics of power spectrum for the region image is carried out. The image is divided into 50 × 50 blocks, and the power spectrum is calculated in each block. The sum of power spectrum of the frequency between 4 and 20 is utilized to judge the flatness of the image. The following figures show the power spectrum distribution of vegetation, field, sea-land junction and sea water in the frequency range of 4-20.

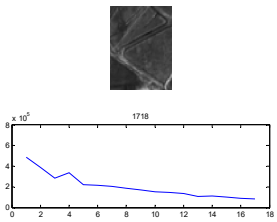


Figure 2.Vegetation power spectrum

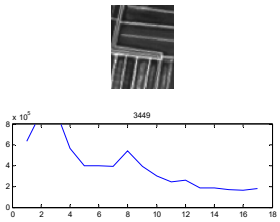


Figure 3.Field power spectrum

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:39:49 UTC from IEEE Xplore. Restrictions apply.

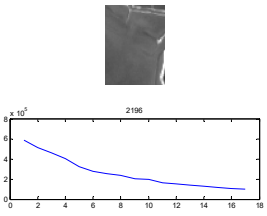


Figure 4. Power spectrum at sea-land junction

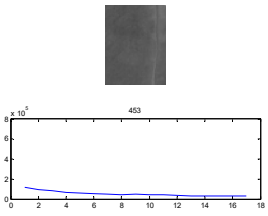


Figure 5. Sea water power spectrum

From the power spectrum of the image, it can be clearly seen that the power spectrum of the image in the sea water is smaller in the area of non-sea water, while in the area that is not seawater in the image, even if it is relatively smooth, the power spectrum of the image is greater. For example, in the field, the power spectrum of the image is significantly greater than that of the seawater. After segmentation, the image can be divided into three categories, the first is the part with higher spectral value, the second is the part with lower spectral value and the third is the part between them. And then we employ canny operator to extract the edge of the intersection of the sea water and the land and extract the edge as the seed point can grow out the region. After the expansion and corrosion algorithm of the image, the interference of the shallow water area is removed. After that the median filter is applied to the binary image and further edge extraction algorithm is used to outline the coastline [5]. The above method realizes the preliminary sea-land segmentation which is the first step to realize the detection and separation of the sea and land areas. Through the preliminary sea-land segmentation, the subsequent detection can be limited to the sea areas where ship targets may appear, reducing the complexity of ship detection and improving the detection efficiency. The above results show that this method is effective for the preliminary segmentation of the infrared remote sensing image.

IV. GIS DATABASE POSITIONING

In order to further reduce the false alarm rate and improve the accuracy of sea-land segmentation in the complex background, the method of combining GIS database positioning to eliminate the land interference is also used in addition to the combination of power spectrum statistics based on blocks and region growing. Because of the obvious strip noise of infrared remote sensing image and the land environment is very complex, which make it difficult to separate the sea and the land accurately. The method of GIS database positioning is to get the land template by the known sea-land database [6]. Then according to the satellite auxiliary data, the acquired image can be located with high precision, and the land area in the acquired image is eliminated by the template. According to the GIS database, the coastline matching is used to separate the sea and the land. The land and the island in the world map are stored in the database in the form of vector map, as shown in the figure below.

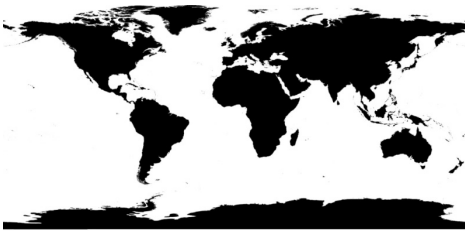


Figure 6. GIS database template

In the method of sea-land segmentation based on GIS template, the images collected on the satellite are first divided into 256 × 256 small blocks, and then the mid-point of each small block image is located to get a certain latitude and longitude. Each small image corresponds to the vector map in the GIS database and can obtain which area belongs to the land or the ocean, which helps to complete the block sea-land segmentation.

The key of this method is to locate the image on the satellite with high precision. Firstly, the relevant parameters of the strict geometric processing model of the optical satellite image are analyzed and the geometric positioning model suitable for on-board processing is constructed by optimizing the parameters. Then, the error analysis and classification are carried out according to the positioning results to find the equivalent parameters of the error source. Then the positioning parameter correction model is conducted with the ground control point data, so as to correct the on-board processing relevant parameters of geometric positioning model which can eliminate system error and partial random error and improve the direct positioning accuracy [7]. After positioning, it is matched and processed with GIS database. The accuracy of sea-land segmentation is improved by GIS database matching, which is beneficial to the subsequent ship target recognition.

The real satellite data are verified by using the positioning algorithm based on the system parameter correction. We select a number of image pixel coordinates from the image source and calculates the corresponding pixel coordinates through the simulation positioning software, and records the direct positioning of the point and the positioning information after the parameter correction with the control points as the reference. The Figure 7 shows the original infrared image contained the land and the sea.



Figure 7. The original infrared image

The following table shows the direct positioning and corrected positioning results of the above infrared image.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:39:49 UTC from IEEE Xplore. Restrictions apply.

Table 1. Direct positioning and correction positioning results

absolute longitude and latitude		Latitude and longitude before correction		Error (m)	Corrected latitude and longitude		Error (m)
38.3727777	117.836765	38.377205	117.840444	588	38.37321	117.836556	52
38.3657875	117.870505	38.369815	117.875127	603	38.36572	117.870797	27
38.3443186	117.794448	38.348085	117.798592	554	38.34394	117.794366	43
38.3241671	117.753317	38.327976	117.757917	584	38.32374	117.753551	52
38.2927609	117.753323	38.296844	117.758811	660	38.29238	117.753725	55

From the above table it can be seen that when we select the several image coordinates randomly, the error of direct positioning result can be controlled within 1000m, and the error of positioning using the modified parameter algorithm can be controlled within 100m, which greatly improves the positioning accuracy and increases the accuracy of sea-land segmentation [8].

V. RESULTS

The combination of the statistics of power spectrum based on block, region growing method and GIS database matching method can improve the sea-land separation effect and reduce the false alarm near the complex background. The following figures showed the results of sea-land segmentation only based on the combination of power spectrum statistics and region growing based on blocks and the results of sea-land segmentation method combined with GIS database template. The results showed that the effect of sea-land segmentation is better by combining GIS database template.



a) original graph b)Preliminary segmentation results c) Combined with GIS database

Figure 8. Comparison of segmentation results

ACKNOWLEDGMENT

The Project was supported by National Natural Science Foundation of China (61605110). We would like to express sincere gratitude to all those who helped us during the study.

REFERENCES

[1] Wang,H.L,Zhu,M.,Lin C.B., and Chen D.B. 2017 .Ship detection in optical remote sensing image based on visual saliency and AdaBoost classifier. *Optoelectronics Letters*.13, 2, 151–155.

[2] Chen J.H.,Liu W.,Zhao Y.J.,and Zeng Y.F.2016. Ship Matching in High-Resolution SAR Imagery.*Journal of Information Engieering University*.17,3,292-297.

[3] Sugimoto M.,Ouchi K.,Nakamura Y. 2013. On the novel use of model-based decomposition in SAR polarimetry for target detection on the sea[J].*Remote Sensing Letters*. 4,9,843-852.

[4] Zhuo Z.M., Miao D.C., and Yang X.Y., 2007.Detection Method for IR Chip Target in Complex Sea Background.*Chinese Journal of Sensors and Actuators*.20,8,1934-1936.

[5] Wang M.G., Wang C., Fan Y.P. 2012.Research For Target Recognition of Infrared Bridge Based on Morphological Operator And Bridge Template.*2012 International Conference on Computer Science and Service System*.58–62.

[6] Yuan X.X.,Cao J.S.,and Yao N. 2009. A Rigorous Geometric Model Considering the Variety of Side Watch Angle for High-resolution Satellite Imagery.*Acta Geodaetica et Cartographica Sinica*.38,2,120-124.

[7] Zhang Y., Wang T., Xu Q., Liu C. J. , and Zhao L. 2006.Rectification for Linear Whiskbroom CCD Imagery Acquired on Pilotless Aircraft .*Journal of Zhengzhou Institute of Surveying and Mapping*.23,3,168-170.

[8] Li F., Zhang H.W.,Wu X.M., and Deng Y.Y. 2007. Aerial Photogrammetry Accuracy Analysis of No Control Points or Less Control Points Based on GPS Differential Technolog. *Chinese Journal of Engineering Geophysics*.4,5,517-519.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:39:49 UTC from IEEE Xplore. Restrictions apply.